

STUDY GUIDE

Neuroanatomy & Developmental Anatomy

A complete, exam-ready revision guide

ANA 252 • NEUROANATOMY

Embryology • Gametogenesis • Fertilization • Weeks 1–3 • Nervous System • PNS • Blood–Brain Barrier & CSF

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How to use this guide: each numbered section builds on the one before it. Key terms are in **navy**, important values and figures in **crimson**, and shaded boxes flag must-know notes and clinical points.

1.1 What Embryology Is, and Why It Matters

Embryology is the study of how a single fertilized cell develops into a complete human being. Human development is a continuous process that begins at **fertilization** — when an oocyte (ovum) from the female is fertilized by a sperm (spermatozoon) from the male — and does not stop at birth.

A coordinated set of processes transforms the fertilized oocyte (the zygote) into a multicellular organism: **cell division, cell migration, programmed cell death, differentiation, growth, and cell rearrangement**. The study of development from the cell stage through organs to the whole body is called **developmental anatomy**.

Significance of studying embryology

- Bridges the gap between prenatal development and obstetrics, paediatrics and clinical anatomy.
- Builds understanding of the beginnings of human life and the changes during prenatal development.
- Helps explain the causes of variation among humans.
- Illuminates gross anatomy and clarifies how normal relationships — and abnormalities (congenital anomalies) — arise.

Why it counts clinically

A central purpose of embryology is to understand congenital anomalies and their causes (etiology), so that such anomalies can be recognised, managed, and where possible prevented.

1.2 A Brief History of Embryology

For centuries the dominant idea was **preformation** — the belief that semen contained a tiny, fully formed infant (a “homunculus”) that simply grew larger. Its supporters, the **preformists**, split into two camps: **ovists** and **spermists**.

Preformation gave way to **epigenesis**, first advanced by **Aristotle**, which held that an animal's form emerges gradually from a relatively formless egg. As microscopy improved in the 19th century, biologists saw embryos take shape in progressive steps, and epigenesis replaced preformation as the favoured explanation.

Contributor	Contribution to embryology
Aristotle	First proposed the theory of epigenesis — form emerges gradually from the egg.
Claudius Galen	Greek physician in Rome; described fetal development and nutrition and the structures now called the allantois, amnion and placenta.
Leonardo da Vinci	Made accurate drawings of the pregnant uterus and introduced a quantitative approach by measuring prenatal growth.
Edwards & Steptoe	Pioneered in-vitro fertilization (IVF); their work led to the birth of Louise Brown in 1978, the first “test-tube baby.”

1.3 Developmental Periods

Human development is customarily divided into a **prenatal** (before birth) and a **postnatal** (after birth) period. Birth itself is a dramatic event that changes the environment, but development does not stop there — important changes continue through infancy, childhood, adolescence and early adulthood (for example, eruption of the teeth and development of secondary sex characteristics).

Period	Span	Key points
Embryonic	Fertilization → end of week 8 (day 56)	The developing human is called an embryo . The most visible advances occur in weeks 3–8.
Fetal	Day 57 → birth	The developing human is called a fetus ; continues until completely outside the mother.
Postnatal	After birth	Growth and maturation continue (teeth, secondary sex characteristics, etc.).

1.4 Key Terminology

Most embryology terms are of Greek or Latin origin and are used by embryologists, obstetricians, gynaecologists and other health professionals to describe the developing human precisely and without ambiguity.

Term	Meaning
Oocyte (ovum, egg)	Female germ/sex cell produced in the ovaries; the mature form is the secondary oocyte.
Sperm (spermatozoon)	Male germ/sex cell produced in the testes (from Greek <i>sperma</i> , “seed”).
Zygote	Cell formed by union of oocyte and sperm at fertilization — the beginning of a new human being.
Gestational age	Age of embryo/fetus from the first day of the last normal menstrual period (LNMP); ~2 weeks longer than fertilization age.
Cleavage	Series of mitotic divisions of the zygote producing early embryonic cells (blastomeres).
Morula	Solid ball of ≥16 blastomeres (Latin <i>morus</i> , mulberry).
Implantation	Attachment of the blastocyst to the endometrium and its subsequent embedding within it.
Embryo / Fetus	Developing human in early stages / the unborn offspring after week 8.
Abortion	Academic term for expulsion of a conceptus before it is viable; may be spontaneous (natural) or induced for medical (therapeutic) reasons.

1.5 Chromosomes, Ploidy and the N Number

Chromosomes are structures in the cell nucleus that carry DNA and allow it to be copied accurately during cell division. **Ploidy** refers to the number of chromosomes in a cell, while the **N number** refers to the amount of DNA in a cell.

- Human somatic (body) cells contain **46 single chromosomes** — the **diploid** number.
- These occur in **23 homologous pairs**: one member of each pair is maternal, the other paternal.

- The 23 pairs comprise **22 pairs of autosomes** plus **1 pair of sex chromosomes**.
- Sex chromosomes **XX** = genetically female; **XY** = genetically male.
- Each gamete carries the **haploid** number of **23 chromosomes**; union at fertilization restores the diploid 46.

Cell renewal categories

Cell type	Behaviour	Examples
Labile	Divide continuously throughout life, replacing lost cells.	Epithelia of ducts, hematopoietic stem cells, epidermis.
Stable	Multiply when needed; otherwise rest in the quiescent G ₀ phase.	Liver, kidney, endocrine glands.
Permanent	Cannot replicate after maturing (post-mitotic).	Skeletal muscle cells, neurons.

2 Cell Division & Gametogenesis

2.1 Meiosis

Meiosis is the special two-division cell division that occurs **only in germ cells** during gametogenesis. It reduces the chromosome number from diploid (2N) to haploid (N), so that diploid germ cells give rise to haploid gametes (sperms and oocytes).

Why meiosis matters in gametogenesis

- Keeps the chromosome number constant across generations by halving it (diploid → haploid).
- Allows random assortment of maternal and paternal chromosomes between gametes — genetic variability.
- Relocates chromosome segments by **crossing over**, shuffling genes and recombining genetic material.

Feature	Meiosis I	Meiosis II
Synapsis	Present	Absent
Crossing over	Present	Absent
Alignment	Homologous pairs align at metaphase plate	23 duplicated chromosomes align at metaphase plate
Disjunction	Homologous chromosomes separate (reduction division)	23 duplicated chromosomes split into single chromosomes; centromeres split
Result	Two haploid cells (chromosomes still duplicated)	Four gametes, each 23 single chromosomes (1N)

Note

DNA is synthesised during the **S phase** of interphase, before meiosis begins. Phosphorylation of histones drives chromosome condensation as division starts.

2.2 Spermatogenesis (Male Gametogenesis)

Spermatogenesis is the sequence of events by which **spermatogonia** are transformed into mature sperms. The primordium of the germ cells reaches the testis at about week 4 and stays dormant until **puberty** (~13–16 years). The whole process takes about **2 months (~64 days)** and continues into old age.

Sequence of events

- At puberty, dormant spermatogonia in the **seminiferous tubules** increase in number by mitosis.
- After several mitotic divisions they grow into **primary spermatocytes** — the largest germ cells in the tubules.
- Each primary spermatocyte undergoes the **1st meiotic (reduction) division** → two haploid **secondary spermatocytes**.
- Each secondary spermatocyte undergoes the **2nd meiotic division** → four haploid **spermatids**.
- Spermatids are remodelled into mature sperms by **spermiogenesis** (a differentiation process, not a division).
- Mature sperms enter the lumen of the seminiferous tubules, then pass to the **epididymis** for storage and functional maturation.

Supporting cell

Sertoli cells lining the seminiferous tubules support and nourish the developing germ cells and help regulate spermatogenesis.

Structure of the mature sperm

A mature sperm is a free-swimming, motile cell with a **head**, **neck** and **tail**. The neck joins head to tail. The head holds the haploid nucleus (23,X or 23,Y); its anterior two-thirds is capped by the **acrosome**, which contains enzymes including **acrosin**.

Tail segment	Feature / function
Middle piece	Contains mitochondria that supply the ATP needed for sperm activity.
Principal piece	Longest segment; provides the main propulsive machinery.
End piece	Terminal segment; contributes to motility toward the fertilization site.

2.3 Oogenesis (Female Gametogenesis)

Oogenesis is the sequence of events by which **oogonia** are transformed into mature oocytes. It begins **before birth**, is completed only after puberty, and continues until **menopause**. Oogenesis has a prenatal and a postnatal phase.

Prenatal maturation

- Primordial germ cells migrate from the wall of the yolk sac and reach the ovary at **week 5**, becoming oogonia (46, 2N).
- Oogonia multiply by mitosis; most then form **primary oocytes**.
- By the **5th month** of prenatal life, germ cells peak at about **7 million**.
- All primary oocytes are formed by the 5th fetal month; cell death (atresia) then begins.
- **No oogonia remain at birth**. Surviving primary oocytes arrest in **prophase of meiosis I**.
- A primary oocyte enclosed by a single layer of flattened follicular cells is a **primordial follicle**.

- **Oocyte maturation inhibitor (OMI)**, secreted by follicular cells, keeps the oocyte arrested until puberty.

Postnatal maturation

- From puberty, usually **one follicle matures each month** and ovulation occurs (unless suppressed by oral contraceptives).
- Primary oocytes at birth: ~**600,000–800,000**; only ~**40,000** remain at puberty; fewer than **500** are ever ovulated.
- As the follicle matures, the primary oocyte completes **meiosis I** → a **secondary oocyte** + the **first polar body**.
- Division of cytoplasm is **unequal**: the secondary oocyte keeps almost all cytoplasm; the small polar body soon degenerates.
- The secondary oocyte enters **meiosis II** but arrests at **metaphase II**, ~3 hours before ovulation.
- Meiosis II completes **only if a sperm penetrates** the oocyte, producing the fertilized oocyte (zygote) + the second polar body.
- If not fertilized, the cell degenerates ~24 hours after ovulation.

2.4 Comparing the Gametes

Feature	Sperm	Oocyte
Motility	Motile (free-swimming)	Immotile
Size	Small	Large / massive
Coverings	No zona pellucida or corona radiata	Surrounded by zona pellucida & corona radiata
Cytoplasm	Sparse	Abundant cytoplasmic granules
Sex chromosome	Two kinds: 23,X or 23,Y	One kind: 23,X
Production	Continuous from puberty (new primary spermatocytes)	No new primary oocytes form after birth

3 Gamete Transport & Fertilization

3.1 Ovarian Events: The Corpus Luteum

Shortly after ovulation the walls of the ovarian follicle and theca folliculi collapse and fold, and are vascularised by surrounding vessels. Under the influence of **luteinizing hormone (LH)** they develop a yellow pigment and become **lutein cells**, forming the **corpus luteum**, which secretes **progesterone** and some **estrogen**. These hormones drive the endometrial glands to secrete and prepare the endometrium for implantation.

If the oocyte is...	What happens to the corpus luteum
Fertilized	Enlarges into the corpus luteum of pregnancy , increasing hormone output. Its degeneration is prevented by human chorionic gonadotropin (hCG) ; it stays active through the first ~20 weeks of pregnancy.
Not fertilized	Involutes and degenerates 10–12 days after ovulation (corpus luteum of menstruation), then becomes white scar tissue, the corpus albicans .

Note

Menopause — the permanent cessation of menstruation — usually occurs between the ages of 48 and 55.

3.2 Oocyte Transport

- At ovulation, the **secondary oocyte** and escaping follicular fluid are expelled from the ovarian follicle.
- The fimbriated end of the uterine tube applies itself closely to the ovary; the **fimbriae** sweep back and forth.
- The sweeping fimbriae plus fluid currents from their cilia carry the oocyte into the funnel-shaped **infundibulum**.
- From the infundibulum the oocyte passes into the **ampulla**, mainly by **peristalsis** of the tubal wall.

3.3 Sperm Transport

- Sperms are produced in the **seminiferous tubules of the testes** and stored in the **epididymis** (mainly its tail).
- Peristaltic contractions of the thick muscular **ductus (vas) deferens** rapidly carry sperms to the urethra.
- Accessory glands — **seminal vesicles, prostate, bulbourethral glands** — add secretions to form semen.
- ~**200–600 million** sperms are deposited around the external os and in the vaginal fornix during intercourse.
- Sperms move through the cervical canal by tail movements; passage through uterus and tubes relies mainly on muscular contractions.

Roles of seminal secretions

Secretion / factor	Role
Vesiculase (seminal glands)	Coagulates part of the ejaculate into a vaginal plug that may prevent backflow of semen.
Prostaglandins	Stimulate uterine motility, assisting sperm movement toward the ampulla.
Fructose (seminal glands)	Energy source for the sperms in semen.
Cervical mucus (at ovulation)	Increases in amount and becomes less viscid — more favourable for sperm transport.

Key figures — the journey

- Ejaculate volume averages **3.5 mL** (range 2–6 mL).
- Sperms move 2–3 mm/min; nonmotile in storage, motile in the ejaculate; slow in the acidic vagina, faster in the alkaline uterus.
- Motile sperms reach the ampulla as early as **5 minutes** after deposition; some take up to **45 minutes**.
- Only about **300–500** sperms reach the fertilization site; the rest degenerate and are resorbed.

3.4 Maturation of Sperm: Capacitation & the Acrosome Reaction

Freshly ejaculated sperms **cannot** fertilize an oocyte. They must first undergo conditioning, in two steps:

Capacitation

Lasting about **7 hours**, capacitation removes the **glycoprotein coat** and **seminal plasma proteins** from the plasma membrane overlying the acrosomal region. Only **capacitated** sperms can pass freely through the corona radiata and undergo the acrosome reaction.

Acrosome reaction

- Occurs after the sperm binds to the zona pellucida. The intact acrosome binds the zona glycoprotein **ZP3**.
- Release of acrosomal enzymes — **hyaluronidase** and **acrosin** — lets the sperm penetrate the zona pellucida.
- The reaction must be completed before the sperm can fuse with the oocyte.
- The sperm plasma membrane, **calcium ions**, **prostaglandins** and **progesterone** all play critical roles.

3.5 Fertilization

The usual site of fertilization is the **ampulla of the uterine tube**; the process takes about **24 hours**. It is a coordinated sequence of stages:

Stage	Event
I. Passage through the corona radiata	Capacitated sperms pass through the corona radiata (capacitation = removal of the glycoprotein coat and seminal plasma proteins).
II. Penetration of the zona pellucida	The acrosome binds zona glycoprotein ZP3 ; acrosin is released, allowing penetration. On contact, cortical-granule lysosomal enzymes alter the zona to block other sperm (zona reaction) — only one sperm penetrates.
III. Fusion of plasma membranes	Oocyte and sperm membranes fuse and break down; the sperm head and tail enter the oocyte cytoplasm while the sperm membrane stays behind.
IV. Completion of meiosis II + female pronucleus	Penetration activates the oocyte to finish meiosis II, forming a mature oocyte and a second polar body. Its nucleus becomes the female pronucleus .
V. Formation of the male pronucleus	Inside the oocyte, the sperm nucleus enlarges into the male pronucleus and the tail degenerates.

Must-know notes

- All sperm mitochondria degenerate — so **all zygote mitochondria (and mtDNA) are of maternal origin**.
- The oocyte with two haploid pronuclei (each 23 chromosomes) is an **ootid**; when the pronuclei fuse into a single diploid set, it becomes a **zygote**.
- The chromosomes then arrange on a cleavage spindle, ready for the first cleavage division.

3.6 Clinical Correlates

Sperm counts & male fertility

- Sperms make up **less than 10%** of semen; the rest is glandular secretion.
- Normal males: usually **>100 million** sperms/mL. ~20 million/mL (or 50 million total) is probably fertile.
- **<10 million/mL** is likely sterile, especially with immotile or abnormal sperms.
- For potential fertility, 50% should be motile after 2 hours and some after 24 hours.
- Male infertility (~30–50% of couple infertility) may follow low count, poor motility, drugs/medications, endocrine disorders, pollutants, smoking, excess alcohol, abnormal sperms, or duct obstruction.

Vasectomy & dispermy

- **Vasectomy** (excision of a segment of each ductus deferens) is the most effective male permanent contraception; reversible in >50% of cases. Afterwards there are no sperms in the ejaculate, but its volume is unchanged.
- **Dispermy**: when two sperms fertilize one oocyte, producing a zygote with an extra set of chromosomes (abnormal).

4 First Week of Development (Days 1–7)

The first week comprises three events: **fertilization** (Part 3), **cleavage and blastocyst formation**, and the **beginning of implantation**.

4.1 Cleavage

- **Cleavage** = repeated mitotic divisions of the zygote, rapidly increasing cell number; the cells are **blastomeres**.
- Blastomeres get **smaller** with each division; cleavage occurs as the zygote travels along the uterine tube toward the uterus.
- Division begins **~30 hours** after fertilization (day 2): 1st cleavage → 2 cells, 2nd → 4 cells, 3rd → 8 cells.

After the 8-cell stage the blastomeres align tightly into a compact ball — **compaction** — probably mediated by cell-surface adhesion glycoproteins. About **3 days** after fertilization the cells divide again to the 16-cell **morula** stage (16–32 blastomeres).

4.2 Blastocyst Formation

- **~4 days** after fertilization, once the morula enters the uterus, a fluid-filled **blastocystic cavity** appears inside it (fluid from the uterine cavity passes through the zona pellucida).
- Fluid separates the blastomeres into the inner **embryoblast** (inner cell mass) and the outer **trophoblast** (outer cell mass).

- By the late blastocyst (day 5) the **zona pellucida degenerates** — a prerequisite for implantation; only then can the blastocyst enlarge considerably.

Note

Because each cleavage daughter cell is smaller than its parent, the embryo does not grow in size until the zona pellucida degenerates.

4.3 Beginning of Implantation

The uterine wall has three layers: the **endometrium** (mucosa, inner lining), the **myometrium** (thick smooth muscle), and the **perimetrium** (outer peritoneal covering). About **6 days** after fertilization ($\approx$ day 20 of a 28-day cycle) the blastocyst attaches to the endometrial epithelium.

- On attachment the trophoblast proliferates and differentiates into an inner **cytotrophoblast** and an outer **syncytiotrophoblast**.
- At $\sim$ day 7 a layer of cells — the **hypoblast** (primary endoderm) — appears on the embryoblast surface facing the blastocystic cavity.

5 Second Week of Development (The Week of Twos)

Hallmarks of week 2: completion of implantation, formation of the **bilaminar embryonic disc** (epiblast + hypoblast), and formation of extraembryonic structures (amniotic cavity, amnion, umbilical vesicle/yolk sac, connecting stalk, chorionic sac).

Day	Key developments
Day 8	Blastocyst partly embedded; syncytiotrophoblast invades endometrium, eroding blood vessels and glands. Embryoblast forms two layers — cuboidal hypoblast (toward blastocyst cavity) and columnar epiblast (toward amniotic cavity), together the bilaminar disc . Amniotic cavity appears in the epiblast; lining cells = amnioblasts .
Day 9	Blastocyst more deeply embedded; surface defect sealed by a fibrin coagulum. Vacuoles in the trophoblast fuse into lacunae (lacunar stage) . Hypoblast cells form the exocoelomic (Heuser's) membrane , lining the primitive yolk sac (primary umbilical vesicle).
Days 11–12	Blastocyst fully embedded. Syncytiotrophoblast erodes endometrial capillaries into sinusoids ; lacunae communicate with sinusoids and maternal blood enters, establishing the primordial uteroplacental circulation . Extraembryonic mesoderm forms; cavities within it coalesce into the extraembryonic coelom (chorionic cavity) .
Day 13	Surface defect fully covered by epithelium (slight implantation-site bleeding can occur). Cytotrophoblast forms cellular columns covered by syncytium — the primary villi . A secondary (definitive) yolk sac forms; pinched-off portions become exocoelomic cysts .

Supporting concepts

- **Extraembryonic somatic mesoderm** lines the cytotrophoblast and amnion; **extraembryonic splanchnic mesoderm** covers the yolk sac.
- The **chorionic plate** = extraembryonic mesoderm lining the inside of the cytotrophoblast.
- The **connecting stalk** is the only place where extraembryonic mesoderm crosses the chorionic cavity; with blood vessels it becomes the **umbilical cord**.

Decidual reaction

As the conceptus implants, endometrial connective-tissue cells swell with glycogen and lipid, becoming **decidual cells**. The decidual reaction provides nutrition for the early embryo and an immunologically privileged site for the conceptus.

6

Third Week of Development (Gastrulation)

Week 3 is characterised by the **primitive streak**, development of the **notochord**, and differentiation of the **three germ layers**.

6.1 Gastrulation & the Three Germ Layers

Gastrulation converts the bilaminar disc into a **trilaminar** disc and is the beginning of **morphogenesis** (development of body form) — the key event of week 3. It begins with the appearance of the primitive streak on the epiblast.

Germ layer	Major derivatives
Ectoderm	Epidermis; central and peripheral nervous systems; the eye; inner ear; neural crest cells.
Mesoderm	Skeletal muscle; blood cells & vessel linings; smooth-muscle coats; serosal linings of body cavities; reproductive & excretory ducts/organs; most of the cardiovascular system; connective tissues — cartilage, bone, tendons, ligaments, dermis, organ stroma.
Endoderm	Epithelial linings of the respiratory and digestive tracts, including glands opening into the gut and glandular cells of organs such as the liver and pancreas.

6.2 The Primitive Streak

- The first sign of gastrulation: epiblast cells proliferate and move to the median plane, forming the streak caudally on the dorsal disc.
- As it elongates, its cranial end forms the **primitive node**; a **primitive groove** develops along the streak, continuous with the **primitive pit** in the node.
- Cells leaving the streak form **mesenchyme** (loosely arranged, amoeboid, phagocytic cells) — the body's supporting connective tissues.
- **Invagination**: epiblast cells move inward — some displace the hypoblast to form **endoderm**, some form **mesoderm**, and those remaining become **ectoderm**. Migration is controlled by **FGF8**.

Fate of the primitive streak

The streak actively forms mesoderm until early week 4, then shrinks to an insignificant structure in the sacrococcygeal region and normally disappears by the end of week 4.

Clinical correlate — Sacrococcygeal teratoma

Remnants of the primitive streak may persist and give rise to a **sacrococcygeal teratoma**. Derived from pluripotent streak cells, it contains tissues from all three germ layers. It is the commonest tumour in newborns (≈ 1 in 35,000), $\sim 80\%$ in females, usually found on antenatal ultrasound, mostly mild; promptly excised with a good prognosis.

6.3 Notochordal Process & the Notochord

- Mesenchymal cells migrate cranially from the primitive node and pit, forming the median **notochordal process**, which acquires a lumen (the **notochordal canal**).
- It grows cranially between ectoderm and endoderm until it reaches the **prechordal plate** — primordium of the **oropharyngeal membrane** (future mouth).
- Caudal to the streak, the **cloacal membrane** marks the future anus. The disc stays bilaminar at both membranes (ectoderm & endoderm fused).
- The **cardiogenic region** lies cranial to the prechordal plate — where the heart primordium develops at the end of week 3.

Functions of the notochord

- Defines the primordial longitudinal axis of the embryo and gives it rigidity.
- Contributes to axial musculoskeletal structures and the central nervous system.
- Contributes to the intervertebral discs.

Formation of the notochord

- The notochordal process elongates by invagination of cells from the primitive pit, which extends in to form the notochordal canal.
- The floor of the notochordal process fuses with the underlying endoderm; the fused layers degenerate, opening the canal to the umbilical vesicle.
- The openings become confluent and the floor disappears, leaving a flattened, grooved **notochordal plate**.

6.4 Clinical Correlates of Early Development

Condition / technique	Key points
Ectopic tubal pregnancy	Blastocyst implants in the uterine tube (delayed transport); the ampulla is the commonest site.
In-vitro fertilization (IVF) & embryo transfer	Oocytes fertilized in vitro and cleaving zygotes transferred to the uterus — helps women with tubal occlusion to conceive.
Intracytoplasmic sperm injection (ICSI)	A single sperm injected directly into a mature oocyte; used when IVF fails or sperm numbers are very low.
Cryopreservation of embryos	Early embryos frozen with a cryoprotectant (e.g. glycerol); 4–8-cell embryos and blastocysts are transferred after thawing.
Surrogate mothers	For women who make oocytes but cannot carry a pregnancy (e.g. after hysterectomy): IVF embryos are transferred to another woman's uterus.

7

Organization of the Nervous System

The nervous system is the body's control and communication network. The brain acts as a central processor; the rest of the system relays messages to and from the brain via the spinal cord and the nerves branching to every organ and body part.

7.1 Two Main Divisions

Division	Composition	Role
Central nervous system (CNS)	Brain + spinal cord	Main centre for correlation and integration of nervous information.
Peripheral nervous system (PNS)	Cranial & spinal nerves + associated ganglia	Relays sensory information to the CNS and carries motor commands out to muscles and glands.

- The CNS is protected by bone — the brain by the skull, the spinal cord by the vertebrae.
- Both are cushioned by membranes called **meninges** and by **cerebrospinal fluid (CSF)**, which protects nerve tissue and removes waste.
- The CNS is built from **neurons** (excitable nerve cells) supported by **neuroglia**; long neuron processes are **axons** (nerve fibres).

7.2 Major Divisions of the Brain

The brain lies in the cranial cavity, continuous with the spinal cord through the **foramen magnum**, surrounded by three meninges — the **dura mater**, **arachnoid mater** and **pia mater** — with CSF in the subarachnoid space. In ascending order from the spinal cord, the brain has three divisions: hindbrain, midbrain, forebrain.

Major division	Subdivisions / parts
Forebrain	Cerebrum + diencephalon (thalamus, hypothalamus, pituitary).
Midbrain	Connects forebrain to hindbrain.
Hindbrain	Medulla oblongata, pons, cerebellum.
Spinal cord	Cervical, thoracic, lumbar, sacral and coccygeal segments.

7.3 The Forebrain

The forebrain is the largest, most complex part. It contains the **cerebrum** — the folded structure that houses intelligence, memory, personality, emotion, speech, and the ability to feel and move — plus deep structures beneath it.

The cerebrum

- Four lobes process different information: **frontal**, **parietal**, **temporal**, **occipital**.
- Two **hemispheres** connected by the **corpus callosum** (a band of nerve fibres).
- The left side is often described as logical/analytical; the right as intuitive/creative.
- The outer layer is the **cortex** ("grey matter"), which receives sensory information and directs it onward.

Deep forebrain structures

Structure	Function
Thalamus	Relays messages from sensory organs (eyes, ears, nose, fingers) to the cortex.
Hypothalamus	Controls pulse, thirst, appetite, sleep patterns and other automatic processes; controls the pituitary gland.
Pituitary gland	Makes hormones controlling growth, metabolism, water/mineral balance, sexual maturity and the stress response.

7.4 Midbrain & Hindbrain

- **Midbrain**: lies beneath the middle of the forebrain; coordinates messages travelling between brain and spinal cord.
- **Hindbrain**: sits beneath the back of the cerebrum and contains the cerebellum, pons and medulla.
- **Cerebellum** (the “little brain”): responsible for balance, movement and coordination.
- **Brainstem** (pons + medulla + midbrain): relays and coordinates messages and controls automatic functions — breathing, heart rate, blood pressure, swallowing, digestion, blinking.

8 The Peripheral Nervous System

The **PNS** comprises all the nerves and ganglia outside the CNS. It relays sensory information from the body to the CNS and carries motor commands from the CNS to muscles and glands.

8.1 Divisions of the PNS

Division	Sub-division	Function
Somatic nervous system	—	Voluntary movement and processing of sensory information.
Autonomic nervous system	Sympathetic	Prepares the body for “fight or flight” responses.
Autonomic nervous system	Parasympathetic	Conserves energy and supports “rest and digest” activities.

Together these systems integrate sensory input, motor responses and autonomic regulation to maintain homeostasis and enable interaction with the environment.

8.2 The Neuron

A **neuron** is the fundamental unit of the nervous system, transmitting electrical and chemical signals and relaying information throughout the body.

Component	Function
Cell body (soma)	Contains the nucleus and most organelles; integrates incoming signals.
Dendrites	Branching extensions that receive signals and relay them to the cell body.
Axon	Long projection carrying action potentials away from the cell body to other cells.
Myelin sheath	Fatty insulating layer for faster transmission — made by Schwann cells (PNS) and oligodendrocytes (CNS).
Nodes of Ranvier	Gaps in the myelin enabling rapid saltatory conduction .
Axon terminals (synaptic boutons)	Release neurotransmitters into the synapse to communicate with the next cell.

8.3 Classification of Neurons

Functional classification (by role)

Type	Role
Sensory (afferent)	Carry signals from sensory receptors (skin, eyes, ears, nose) to the CNS; detect touch, temperature, pain.
Motor (efferent)	Carry commands from the CNS to muscles and glands, enabling movement and responses.
Interneurons (association)	Within the CNS; connect sensory and motor neurons and process information.

Structural classification (by shape)

Type	Features & location
Multipolar	Most common; multiple dendrites + one axon. Brain, spinal cord, motor pathways.
Bipolar	One dendrite + one axon. Special sensory organs — retina, olfactory system.
Unipolar (pseudounipolar)	Single extension branching in two. Sensory neurons for touch and pain.
Anaxonic	No distinguishable axon. Process information and regulate activity in the brain.

8.4 Nerves

Nerves are bundles of axons (nerve fibres) that transmit impulses between the CNS and the body. The PNS has two main types:

Nerve type	Origin	Number & notes
Cranial nerves	Arise from the brainstem	12 pairs ; extend to head, neck and trunk; sensory, motor or mixed functions.
Spinal nerves	Arise from the spinal cord	31 pairs ; exit through intervertebral foramina; named by level — cervical, thoracic, lumbar, sacral, coccygeal.

9.1 The Blood–Brain Barrier (BBB)

The **blood–brain barrier** is a highly selective, semipermeable membrane that separates circulating blood from the brain's extracellular fluid. It protects the brain from harmful substances while letting essential nutrients in, maintaining brain homeostasis.

Components of the BBB

Component	Key features
Endothelial cells	Tightly joined by tight junctions (block paracellular transport); lack fenestrations; low pinocytic activity; can release inactivating enzymes for neurotoxins.
Basement membrane	Thin extracellular-matrix layer giving structural support; contains pericytes.
Astrocyte end-feet	Glial end-feet cover >90% of the capillary surface; secrete factors (e.g. glial-derived neurotrophic factor) that maintain BBB properties.
Pericytes	Embedded in the basement membrane; regulate blood flow, vascular stability, BBB integrity, angiogenesis and immune responses.

How substances cross

The BBB is lipid-based, so it repels water-soluble substances; large molecules cannot pass between the interlocking endothelial cells. Only small, lipid-soluble substances cross easily.

Functions of the BBB

- **Selective permeability:** allows essential molecules (glucose, amino acids) via specific transporters; restricts toxins, pathogens and large/hydrophilic molecules.
- **Homeostasis:** keeps a constant environment — regulating ion balance, neurotransmitter levels and pH.
- **Protection:** shields the brain from blood-borne neurotoxins and blocks peripheral immune cells from entering the CNS.
- **Metabolic role:** endothelial enzymes (e.g. monoamine oxidases) degrade harmful substances.

Transport mechanisms

Mechanism	What it moves
Passive diffusion	Small lipid-soluble molecules — O_2 , CO_2 — down their concentration gradient.
Carrier-mediated transport	Glucose and amino acids via selective carrier proteins.
Active transport	Ions and vitamins against a gradient; requires ATP.
Receptor-mediated endocytosis	Large biomolecules such as insulin and transferrin, via receptor binding.

Circumventricular organs (weak / absent BBB)

Area postrema (vomiting centre), **pineal gland**, **median eminence** and **neurohypophysis** (posterior pituitary) have extensive, highly permeable capillaries.

Clinical relevance

- BBB disruption: infections (meningitis), trauma, ischemia and stroke, multiple sclerosis (immune-cell infiltration), and tumours (leaky angiogenic capillaries).
- Drug delivery is hard for large/hydrophilic drugs — strategies include lipid-soluble drugs, carrier systems (e.g. nanoparticles) and intrathecal/intraventricular delivery.

9.2 Cerebrospinal Fluid (CSF)

CSF is a clear, colourless, plasma-like fluid that bathes the CNS, with a volume of about **80–150 mL** in adults. It circulates through cavities within the brain and spinal cord and is produced mainly by the **choroid plexuses**.

Production

- Produced by the **choroid plexus** in the walls of the lateral ventricles and the roofs of the third and fourth ventricles.
- **Ependymal cells** of the choroid plexus secrete CSF by active transport and ultrafiltration of blood plasma.
- CSF is not a simple ultrafiltrate — it differs from blood in electrolyte, glucose and protein content.

Blood supply to the choroid plexus

Location	Arterial supply
Lateral & third ventricles (anterior)	Anterior choroidal artery (branch of the internal carotid artery).
Lateral & third ventricles (posterior)	Posterior lateral and medial choroidal arteries (branches of the posterior cerebral artery).
Fourth ventricle	Inferior cerebellar arteries.

Circulation of CSF

CSF flows through the ventricular system in order: **lateral ventricles** → (via the **interventricular foramen of Monro**) → **third ventricle** → (via the **cerebral aqueduct of Sylvius**) → **fourth ventricle** → out to the subarachnoid space via the **median aperture (foramen of Magendie)** and the **lateral apertures (foramina of Luschka)**.

Absorption

CSF is absorbed at the **arachnoid granulations** along the superior sagittal sinus, diffusing into the dural venous sinuses and cerebral veins. The granulations act as a one-way valve — allowing CSF into the bloodstream but preventing backflow of blood into the CSF.

Functions of CSF

Function	Detail
Mechanical protection	Cushions / absorbs shock for the brain and spinal cord, preventing mechanical injury.
Buoyancy	The brain floats in CSF, cutting its effective weight from ~1,400 g to ~50 g.
Chemical stability	Maintains CNS homeostasis, removes waste and provides the ionic balance needed for action potentials.
Nutrient transport	Delivers glucose and ions and distributes neuroactive substances and hormones.

Clinical relevance

Topic	Key points
Lumbar puncture (spinal tap)	Performed at L3–L4 or L4–L5. Used to diagnose meningitis (bacterial, viral, fungal), MS and tumours, to measure CSF pressure, and to give spinal anaesthesia or intrathecal drugs.
Hydrocephalus	Abnormal CSF accumulation in the ventricles. Communicating = impaired reabsorption; non-communicating = blockage in the ventricular pathway; normal-pressure = elderly, triad of gait disturbance, dementia and urinary incontinence.
CSF leaks	From trauma, surgery or spontaneous causes; can cause intracranial hypotension and recurrent meningitis.
Intracranial pressure (ICP)	Raised in tumours and haemorrhages; low after leaks or lumbar puncture.

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